

resulting strong adverse pressure gradient by a strong shock wave combined with the low velocity of the boundary layer results in separation and reattachment of the boundary layer, thereby greatly increasing the complexity of the flowfield. The cold-wall effect of the hypersonic flow may significantly narrow the interference region of the shock wave/boundary-layer interaction [13], which further increases the gradient of the flow variables and physical properties near the wall; and it reduces the accuracy of prediction on the flowfield and surface aerothermodynamic characteristics.

Due to the high-Mach-number and high-enthalpy characteristics of hypersonic flows, the cost and difficulty of hypersonic experiments have increased significantly as compared to regular wind-tunnel experiments [14]. However, with the development of computer science, the ability of computational fluid dynamics (CFD) to solve engineering problems is rapidly improving [15,16]. The most complex aspect of CFD is the treatment of turbulent flows, and there are mainly three numerical paradigms for dealing with turbulence: direct numerical simulation (DNS) [17], large-eddy simulation [18], and Reynolds-averaged Navier–Stokes (RANS) equations [19,20]. The RANS method is the most widely applied solution to CFD engineering problems because of its minimal computing resources required. Therefore, its low cost and ease of implementation have made it one of the most important tools for high-speed aerospace design [15,21]. The turbulence simulation of RANS relies heavily on selecting and tuning turbulence models. Among them, the eddy viscosity models are relatively simple and well developed, which can be broadly classified as zero-equation, one-equation, and two-equation models. The zero-equation model is simple in form but ignores the anisotropy of Reynolds stress and the historical effects of turbulence, whereas the one-equation model takes further account of the historical effects of turbulence and the two-equation model incorporates the scaling characteristics of turbulence. However, either turbulence model was developed for incompressible flows, which raised questions about the applicability and accuracy of turbulence models in the case of high-Mach-number flows with strong compressibility effects. In related works, Rumsey [22] and Aiken et al. [23] observed that the accuracy of existing turbulence models for predicting heat flux and friction coefficients in hypersonic flows decreases with the increase of cold-wall effects and the Mach number.

Focusing on eddy viscosity models, Roy and Blottner [24] and Celic and Hirschel [25] comprehensively reviewed several widely used turbulence models and found that the Spalart–Allmaras (SA) one-equation model [26] and the shear-stress transport (SST) $k - \omega$ two-equation model [27] were usually able to give accurate predictions in the models they studied [28–30]. In the assessment and quantification of SA models, Coratekin et al. [31] compared the simulation performances of different turbulence models and different upwind schemes under hypersonic flows and found that the original SA turbulence model had the most accurate prediction of the pressure profile when the compression effect was stronger. Erb and Hosder [32] performed uncertainty quantification and sensitivity analysis on the original SA model and the Wilcox $k - \omega$ model [33] for the hypersonic shock wave/boundary-layer interaction problems. Li et al. [34] performed Bayesian uncertainty quantification and sensitivity analysis on the original SA model, the Quadratic Constitutive Relation correction SA model (SA-QCR2000) proposed in 2000 by Spalart model [35], and the SST $k - \omega$ two-equation model in hypersonic shock wave/boundary-layer interaction problems; and they reported that the peak uncertainty of the SA-QCR2000 model was smallest after enhanced by Bayesian estimation. Lagares-Nieves et al. [36] compared the predictions of first-order statistics obtained by the standard compressible SA model, the Wilcox $k - \omega$ model, and the SST $k - \omega$ model in concave wall boundary-layer cases and found that the SA model was the best compromise between accurate predictions and numerical stability. Taking into account the fast computational speed, the robustness, and the simplicity of the SA model itself, the main research object of this paper will be the SA turbulence model.

The SA turbulence model was first proposed by Spalart and Allmaras [26] as a one-equation model based on the eddy viscosity transport equation, which directly assumes that the turbulent viscosity coefficients satisfy the scalar equations in the flowfield and gives empirical

constants and empirical functions based on experience. These empirical constants and empirical functions are well tuned due to the extensive experience and rich experimental calculation data in regular aeronautical calculations. The first extension of the SA model to hypersonic flows was derived in 1996 when Edwards and Chandra [37] pointed out that sufficiently large inflow values of turbulence variables overwhelmed the effect of the f_{r2} term in the standard SA model, and so the f_{r2} term in the original model was removed and some of the parameters were redefined. Its convergence was further improved by improving the near-wall numerical behavior; then, its accuracy in profiles and surface coefficients prediction was tested under high-Mach-number arithmetic. As for the work devoted to compressibility corrections of the SA model, Shur et al. [38], Catris [39], Catris and Aupoix [40], Séror et al. [41], and others have further improved the accuracy of the model in compressible flows by adding source or diffusion terms to the standard SA equation. In 2012, Allmaras and Johnson [42] provided a standardized description of the compressible correction of the SA model (hereafter referred to as the SA-2012); in addition to this, they proposed the negative Spalart–Allmaras model to address issues with under-resolved grids and nonphysical transient states in discrete settings, and they corrected some of the coefficients in the original SA equations to prevent the turbulence variables from becoming negative. In terms of hypersonic applications, Tu et al. [43] analyzed the two compressible correction forms proposed by Catris [39,40] in combination with the correction proposed by Shur et al. [38] in high-Mach-number corner cases and found that the correction model proposed by Catris and Aupoix [40] (SA-Catris-2000) superimposed on the correction proposed by Shur et al. [38] (SA-COMP) and Catris's revised model (SA-Catris-1999) [39] are both more accurate in predicting pressure profiles and heat flux profiles than the standard compressible SA model, and the former one gives a better improvement. Gnoffo et al. [44] and Kumar and Nair [45] validated the SA-Catris-2000 compressibility correction model with different high-Mach-number compressional corner cases and found that it could predict the pressure profiles and heat flux profiles more accurately than the original model. Recently, Waligura et al. [46] improved the compressibility correction model proposed by Catris and Aupoix in 2000 [40], explored the nonconservation form of the SA equation, and evaluated its ability to predict heat flux at high-Mach-number flows; they found that the SA model in conservation form predicted heat flux better than the SA-2012 model [42], the SA-QCR2000 model, and the SA-Catris-2000 model. Aiken et al. [23], on the other hand, pointed out that the SA-Catris-2000 model with the low-Reynolds-number correction [47] predicted better than the SST $k - \omega$ model for flat-plate profiles at high-Mach-number cold-wall effects.

Most of the assessment of the SA turbulence model's compressibility corrections in high-Mach-number flows has been conducted on the SA-Catris-2000 correction model [23,43–46,48]; the performance of other correction forms and the combined effects of the different corrections need to be effectively evaluated in hypersonic flows. Therefore, our work will further summarize the compressibility corrections of the SA turbulence model and assess the accuracy of the different corrections and their combined effects by using the hypersonic flows over a flat plate and a compression corner. The accuracy of different models is compared by plotting error profiles for the flat-plate cases, whereas the coarse-grained uncertainty metrics proposed by Gnoffo et al. [49] is used to measure the accuracy of different models for the corner cases.

The rest of the paper is organized as follows: Sec. II introduces and summarizes the different compressibility corrections for the SA turbulence model. Section III presents the procedures, cases, and error quantification methods used in this paper. The results of the calculations are presented and briefly summarized in Sec. IV. Finally, Sec. V summarizes the whole paper and draws conclusions.

II. Development of Compressible SA Turbulence Models

The commonly used compressibility correction terms of the SA model are introduced in this section. In order to facilitate subsequent programming and comparison, these SA models will be described uniformly.

A. Original SA Model

$$\frac{\partial \tilde{v}}{\partial t} + u_j \frac{\partial \tilde{v}}{\partial x_j} = \underbrace{c_{b1}(1-f_{t2})\tilde{S}\tilde{v}}_{\text{Production}} - \underbrace{\left[c_{w1}f_w - \frac{c_{b1}}{\kappa^2}f_{t2} \right] \left(\frac{\tilde{v}}{d} \right)^2}_{\text{Destruction}} + \underbrace{\frac{1}{\sigma} \left[\frac{\partial}{\partial x_j} \left((\nu + \tilde{v}) \frac{\partial \tilde{v}}{\partial x_j} \right) + c_{b2} \frac{\partial \tilde{v}}{\partial x_j} \frac{\partial \tilde{v}}{\partial x_j} \right]}_{\text{Diffusion}} + \underbrace{f_{t1}\Delta U^2}_{\text{Trip}} \quad (1)$$

The original SA model by Spalart and Allmaras [26] solves the model equation for the Reynolds stress $-u_i' u_j'$ through the turbulence variable $\tilde{v}$, denoting the kinematic turbulent eddy viscosity ν_t . The specific relationship between them is given in Eq. (2):

$$\overline{u_i' u_j'} = \nu_t \left(\frac{\partial \overline{u_i}}{\partial x_j} + \frac{\partial \overline{u_j}}{\partial x_i} \right), \quad \nu_t = \tilde{v} f_{v1}, \quad f_{v1} = \frac{\chi^3}{\chi^3 + c_{v1}^3}, \quad \chi = \frac{\tilde{v}}{v} \quad (2)$$

The first term on the right-hand side of Eq. (1) is the production term; the second term is the destruction term, which describes the transformation of turbulent energy in the flow; the third term is the diffusion term; and the fourth term is the trip term, which is used to adjust the location of turbulence appearance in the flowfield. Considering that the subsequent cases in this paper are all fully turbulent cases and do not make an in-depth study on the transformation of laminar and turbulent flows, all subsequent SA models do not include the trip term. Other specific parameters are determined empirically and were clearly defined in the original references.

B. Standard Compressible SA and SA-2012

The SA model presented in Eq. (1) was constructed for incompressible turbulent flows. To enhance its applicability and validity to cases involving compressible flows, the standard compressible modified version of the SA model can be straightforwardly derived by coupling Eq. (1) with the continuity equation: $\rho \cdot \text{SA} + \tilde{v} \cdot \text{mass} = 0$:

$$\frac{\partial \rho \tilde{v}}{\partial t} + \frac{\partial \rho \tilde{v} u_j}{\partial x_j} = \rho c_{b1}(1-f_{t2})\tilde{S}\tilde{v} - \rho \left[c_{w1}f_w - \frac{c_{b1}}{\kappa^2}f_{t2} \right] \left(\frac{\tilde{v}}{d} \right)^2 + \underbrace{\frac{1}{\sigma} \left[\frac{\partial}{\partial x_j} \left(\rho(\nu + \tilde{v}) \frac{\partial \tilde{v}}{\partial x_j} \right) + \rho c_{b2} \frac{\partial \tilde{v}}{\partial x_j} \frac{\partial \tilde{v}}{\partial x_j} \right]}_{\text{standard}} - \underbrace{\frac{1}{\sigma} (\nu + \tilde{v}) \frac{\partial \rho}{\partial x_j} \frac{\partial \tilde{v}}{\partial x_j}}_{\text{SA-2012}} \quad (3)$$

Equation (3) was initially proposed by Allmaras and Johnson [42] in 2012, which was derived mathematically by incorporating mass conservation equation with the standard SA equation. A cross term was then obtained in the diffusion term, as given in the last term on the right-hand side of Eq. (3). The term was neglected in most of the compressible extensions of the SA model [39–41], whereas it was considered in Refs. [43,46]. The specific effect of the correction term needs to be further verified in high-Mach-number flows.

C. SA-COMP Correction

Shur et al. [38] added a source term to the right-hand side of the original SA equation in order to improve the SA's behavior in compressible mixing layers. The specific expression is presented in Eq. (4), where $C_5 = 3.5$:

$$-C_5 \frac{\tilde{v}^2}{a^2} \frac{\partial u_i}{\partial x_j} \frac{\partial u_j}{\partial x_i} \quad (4)$$

SA-COMP

The correction term can also be applied to the standard compressible SA turbulence model by simply multiplying the term with the density. Tu et al. [43] contributed some preliminary exploration of the

application of Shur et al.'s [38] compressibility correction term to the compressible SA model in 2012. Likewise, the correction term is further evaluated in the current study.

D. SA-Catris-1999 Correction and Séror Compressibility Correction

Because the density gradient is particularly important in hypersonic flows, in 1999, Catris [39] proposed to replace part of the $\tilde{v}$ originally located in the standard compressible SA turbulence model's diffusion term by $\rho \tilde{v}$; the specific expression of the diffusion term in the modified form is given in Eq. (5):

$$\frac{1}{\sigma} \left[\frac{\partial}{\partial x_j} \left(\rho(\nu + \tilde{v}) \frac{\partial \tilde{v}}{\partial x_j} \right) + c_{b2} \frac{\partial \tilde{v}}{\partial x_j} \frac{\partial \rho \tilde{v}}{\partial x_j} \right] = \underbrace{\frac{1}{\sigma} \left[\frac{\partial}{\partial x_j} \left(\rho(\nu + \tilde{v}) \frac{\partial \tilde{v}}{\partial x_j} \right) + \rho c_{b2} \frac{\partial \tilde{v}}{\partial x_j} \frac{\partial \tilde{v}}{\partial x_j} \right]}_{\text{Standard}} + \underbrace{\frac{c_{b2}}{\sigma} \tilde{v} \frac{\partial \rho}{\partial x_j} \frac{\partial \tilde{v}}{\partial x_j}}_{\text{SA-Catris-1999}} \quad (5)$$

Equation (5) shows that the correction form is equivalent to adding a diffusion term

$$\frac{c_{b2}}{\sigma} \tilde{v} \frac{\partial \rho}{\partial x_j} \frac{\partial \tilde{v}}{\partial x_j}$$

to the standard compressible SA turbulence model. Séror et al. [41] further modified the diffusion term in order to resolve the instability issue of the diffusion term in the standard compressible model, where the specific expression for the diffusion term is given in Eq. (6):

$$\frac{1}{\sigma} \left[(1 + c_{b2}) \frac{\partial}{\partial x_j} \left(\rho(\nu + \tilde{v}) \frac{\partial \tilde{v}}{\partial x_j} \right) - c_{b2} \rho(\nu + \tilde{v}) \frac{\partial}{\partial x_j} \frac{\partial \tilde{v}}{\partial x_j} \right] = \underbrace{\frac{1}{\sigma} \left[\frac{\partial}{\partial x_j} \left(\rho(\nu + \tilde{v}) \frac{\partial \tilde{v}}{\partial x_j} \right) + \rho c_{b2} \frac{\partial \tilde{v}}{\partial x_j} \frac{\partial \tilde{v}}{\partial x_j} \right]}_{\text{Standard}} + \underbrace{\frac{c_{b2}}{\sigma} \tilde{v} \frac{\partial \rho}{\partial x_j} \frac{\partial \tilde{v}}{\partial x_j} + \frac{c_{b2}}{\sigma} \frac{\partial \rho \nu}{\partial x_j} \frac{\partial \tilde{v}}{\partial x_j}}_{\text{SA-Séror}} \quad (6)$$

It can be seen in Eq. (6) that Séror et al.'s correction form [41] is equivalent to adding two diffusion terms to the standard compressible model:

$$\frac{c_{b2}}{\sigma} \tilde{v} \frac{\partial \rho}{\partial x_j} \frac{\partial \tilde{v}}{\partial x_j} + \frac{c_{b2}}{\sigma} \frac{\partial \rho \nu}{\partial x_j} \frac{\partial \tilde{v}}{\partial x_j}$$

the former one of the two diffusion terms is the same as the correction term added by SA-Catris-1999 model, and the latter one is the other correction term to be further tested in this work.

E. SA-Catris-2000 Correction

Catris and Aupoix [40] proposed a new correction to the diffusion term in order to make the turbulence model more consistent with the logarithmic law of the compressible boundary layer, where the specific expression for the diffusion term is shown in Eq. (7):

$$\frac{1}{\sigma} \left[\frac{\partial}{\partial x_j} \left(\rho \nu \frac{\partial \tilde{v}}{\partial x_j} \right) + \frac{\partial}{\partial x_j} \left(\sqrt{\rho \tilde{v}} \frac{\partial \sqrt{\rho \tilde{v}}}{\partial x_j} \right) + c_{b2} \frac{\partial \sqrt{\rho \tilde{v}}}{\partial x_j} \frac{\partial \sqrt{\rho \tilde{v}}}{\partial x_j} \right] \quad (7)$$

The preceding equation leads to programming complexity when performing programming calculations due to the presence of $\sqrt{\rho}$ in it, and Waligura et al. [46] introduced the substitution

$$\frac{\partial \sqrt{\rho}}{\partial x_j} = \frac{1}{2\sqrt{\rho}} \frac{\partial \rho}{\partial x_j}$$

into Eq. (7):

$$\begin{aligned}
& \frac{1}{\sigma} \left[\frac{\partial}{\partial x_j} \left(\rho \tilde{\nu} \frac{\partial \tilde{\nu}}{\partial x_j} \right) + \frac{\partial}{\partial x_j} \left(\sqrt{\rho \tilde{\nu}} \frac{\partial \sqrt{\rho \tilde{\nu}}}{\partial x_j} \right) + c_{b2} \frac{\partial \sqrt{\rho \tilde{\nu}}}{\partial x_j} \frac{\partial \sqrt{\rho \tilde{\nu}}}{\partial x_j} \right] \\
&= \frac{1}{\sigma} \left[\frac{\partial}{\partial x_j} \left(\rho(\nu + \tilde{\nu}) \frac{\partial \tilde{\nu}}{\partial x_j} \right) + \rho c_{b2} \frac{\partial \tilde{\nu}}{\partial x_j} \frac{\partial \tilde{\nu}}{\partial x_j} \right] \\
&\quad \underbrace{\hspace{10em}}_{\text{Standard}} \\
&+ \underbrace{\frac{c_{b2}}{\sigma} \tilde{\nu} \frac{\partial \rho}{\partial x_j} \frac{\partial \tilde{\nu}}{\partial x_j} + \frac{1}{\sigma} \frac{\partial}{\partial x_j} \left(\frac{\tilde{\nu}^2}{2} \frac{\partial \rho}{\partial x_j} \right) + \frac{c_{b2}}{\sigma} \frac{1}{4} \frac{\tilde{\nu}^2}{\rho} \frac{\partial \rho}{\partial x_j} \frac{\partial \rho}{\partial x_j}}_{\text{SA-Catris-1999}} \\
&\quad \underbrace{\hspace{10em}}_{\text{SA-Catris-2000}}
\end{aligned} \tag{8}$$

Equation (8) shows that the correction model is equivalent to adding three diffusion terms to the standard compressible model:

$$\frac{c_{b2}}{\sigma} \tilde{\nu} \frac{\partial \rho}{\partial x_j} \frac{\partial \tilde{\nu}}{\partial x_j} + \frac{1}{\sigma} \frac{\partial}{\partial x_j} \left(\frac{\tilde{\nu}^2}{2} \frac{\partial \rho}{\partial x_j} \right) + \frac{c_{b2}}{\sigma} \frac{1}{4} \frac{\tilde{\nu}^2}{\rho} \frac{\partial \rho}{\partial x_j} \frac{\partial \rho}{\partial x_j}$$

The first one is the same as the one added by the SA-Catris-1999 model, and the last two are the other two terms that need to be further explored in this paper. It is worth noting that the correction form has also been used in combination with other correction forms in literature; i.e., Tu et al. [43] combined the correction term with the SA-COMP correction term and examined the performance of that in hypersonic flat-plate cases. S  r  r and Kosarev [48] combined the correction model with the SA-S  r  r model and applied it extensively in a large number of hypersonic cases.

F. Conservation Correction of Compressible SA Model

It is worth noting that, in Refs. [39,40,43], the left-hand sides of the compressible SA equations are indicated as

$$\frac{D\rho\tilde{\nu}}{Dt}$$

whereas they are indicated as

$$\frac{\partial \rho \tilde{\nu}}{\partial t} + \frac{\partial \rho \tilde{\nu} u_j}{\partial x_j}$$

in Refs. [42,48,49], and the specific relationship between them is presented in Eq. (9):

$$\frac{\partial \rho \tilde{\nu}}{\partial t} + \frac{\partial \rho \tilde{\nu} u_j}{\partial x_j} - \underbrace{\rho \tilde{\nu} \frac{\partial u_j}{\partial x_j}}_{\text{nonconservative}} = \frac{D\rho\tilde{\nu}}{Dt} \tag{9}$$

This correction term is known as the nonconservative term (SA-Noncon), and the effect of this term in the high-Mach-number inflection case was explored by Waligura et al. [46], who found that the presence of this term has a large impact on the prediction accuracy of the model; and by Gno  o et al. [49], who also concluded that the presence of this term is detrimental to the SA model in compressible form. However, there are still some papers that retain this term. Therefore, this term is then investigated in our study.

G. Summary of Corrections to the Compressible SA Model

Considering all the compressibility corrections described in the previous six subsections together, here, we simply combine them in a unified equation:

$$\begin{aligned}
& \frac{\partial \rho \tilde{\nu}}{\partial t} + \frac{\partial \rho \tilde{\nu} u_j}{\partial x_j} = \rho c_{b1} (1 - f_{t2}) \tilde{S} \tilde{\nu} - \rho \left[c_{w1} f_w - \frac{c_{b1}}{\kappa^2} f_{t2} \right] \left(\frac{\tilde{\nu}}{d} \right)^2 \\
& + \underbrace{\frac{1}{\sigma} \left[\frac{\partial}{\partial x_j} \left(\rho(\nu + \tilde{\nu}) \frac{\partial \tilde{\nu}}{\partial x_j} \right) + \rho c_{b2} \frac{\partial \tilde{\nu}}{\partial x_j} \frac{\partial \tilde{\nu}}{\partial x_j} \right]}_{\text{standard}} \\
& + \underbrace{\beta_1 \rho \tilde{\nu} \frac{\partial u_j}{\partial x_j}}_{\text{nonconservative}} + \underbrace{\beta_2 \left(\frac{1}{\sigma} \frac{\partial}{\partial x_j} \left(\frac{\tilde{\nu}^2}{2} \frac{\partial \rho}{\partial x_j} \right) + \frac{c_{b2}}{\sigma} \frac{1}{4} \frac{\tilde{\nu}^2}{\rho} \frac{\partial \rho}{\partial x_j} \frac{\partial \rho}{\partial x_j} \right)}_{\text{SA-Catris-2000}} \\
& + \underbrace{\beta_3 \frac{c_{b2}}{\sigma} \tilde{\nu} \frac{\partial \rho}{\partial x_j} \frac{\partial \tilde{\nu}}{\partial x_j}}_{\text{SA-Catris-1999}} - \underbrace{\beta_4 \rho C_5 \frac{\tilde{\nu}^2}{a^2} \frac{\partial u_i}{\partial x_j} \frac{\partial u_i}{\partial x_j}}_{\text{SA-COMP}} \\
& - \underbrace{\beta_5 \frac{1}{\sigma} (\nu + \tilde{\nu}) \frac{\partial \rho}{\partial x_j} \frac{\partial \tilde{\nu}}{\partial x_j}}_{\text{SA-2012}} + \underbrace{\beta_6 \frac{c_{b2}}{\sigma} \frac{\partial \rho \nu}{\partial x_j} \frac{\partial \tilde{\nu}}{\partial x_j}}_{\text{SA-S  r  r}}
\end{aligned} \tag{10}$$

By changing the six β to zero or one in Eq. (10), the different SA turbulence models and combined models can be formed to calculate the hypersonic cases. From the previous derivation process, it can be seen that if β_2 or β_3 is one in Eq. (10), β_6 must also be one; and the remaining terms can be freely combined.

III. Cases and Uncertainty Quantification

A. Numerical Solver and Test Cases

A compressible RANS solver with a multiblock structural mesh based on the finite volume method (named OpenCFD-EC by Li et al.) [50] is adopted as a basic solver. The accuracy of the code has been extensively verified [51]. In this paper, secondary development of the turbulence model part based on OpenCFD-EC is performed by incorporating various SA models, and RANS equations are numerically solved by using the LU-SGS (which stands for Lower-Upper Symmetric Gauss-Seidel) implicit time advance and the MUSCL (which stands for Monotone Upstreamcentered Schemes for Conservation Laws) scheme with the van Albada limiter.

Two types of hypersonic flow cases are selected to evaluate and quantify the turbulence models, where the hypersonic flat-plate case was validated by accurate DNS data on hypersonic flat plates [8,52], and the hypersonic corner case was validated by experimental data summarized by NASA on two-dimensional compressible corner cases [53]. The grid and boundary diagrams of the cases are shown in Figs. 1 and 2. The height of the first layer of the grid for the flat cases is $y^+ \approx 0.1$, and the scale of the flat grid is 245×97 . The corner consists of ranges of the precompression region at $0 < x < 0.5575$ and the postcompression region at $L = 0.254$ m. The grid convergence study has been performed and verified on the meshes. In the final results of the corner case, the height of the first layer of the grid for the postcompression region is chosen to be $y^+ \approx 0.3$ and the grid resolution 159×100 of the corner is adopted.

The relevant inflow conditions for the flat-plate and corner cases are shown in Tables 1 and 2, where the gas for the M6Tw076 case and

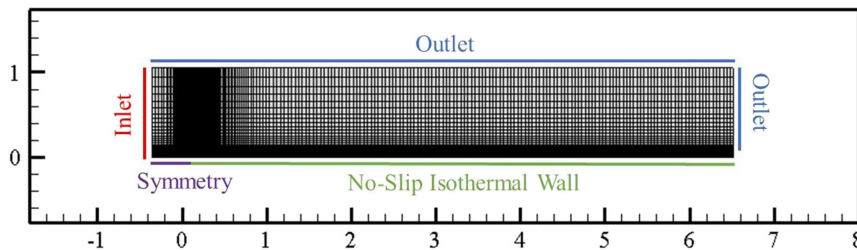


Fig. 1 Schematic diagram of the grid and boundaries of the flat-plate cases.

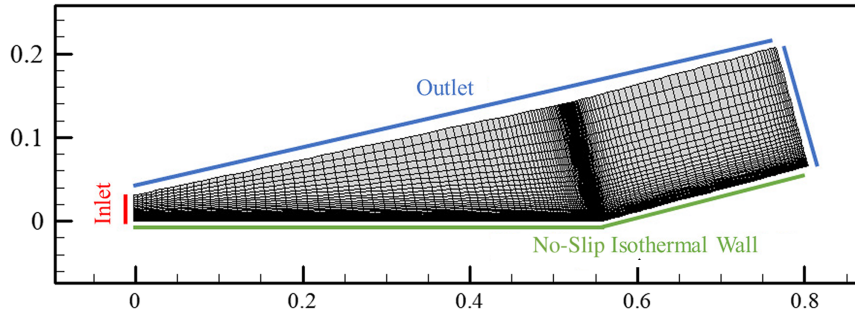


Fig. 2 Schematic diagram of the grid and boundaries of the corner cases.

Table 1 Inflow conditions of flat-plate cases

	M_∞	U_∞ , m/s	ρ_∞ , kg/m ³	T_∞ , K	T_w , K	T_w/T_r	$h_{0,\infty}$, kJ/kg
M6Tw076	5.86	870.4	0.043	55.0	300	0.76	71.5
M6Tw025	5.84	869.1	0.044	55.2	97.5	0.25	72.1
M8Tw048	7.87	1155.1	0.026	51.8	298	0.48	71.2

Table 2 Inflow conditions of corner cases of flat-plate cases

θ deg	M_∞	Re , m ⁻¹	T_0 , K	T_∞ , K	T_w , K	$Q_{\text{exp},\infty}$, W/m ²
15	9.22	4.7e-7	1070	64.5	295	6.07e-4
34	9.22	4.7e-7	1070	64.5	295	6.29e-4

the M6Tw025 case is air and the gas for the remaining cases is nitrogen; this is mainly because, when the Mach number of air is greater than six, the effect of chemical reactions cannot be negligible.

B. Coarse-Grained Uncertainty Metrics

Considering the large amount of data and models involved in this paper, a suitable error metric needs to be considered to appropriately measure the strengths and weaknesses of different models. Owing to the huge database requirement of the Monte Carlo method [54] and the nonintrusive probabilistic collocation [55], the coarse-grained uncertainty metrics are chosen for the error analysis of the calculations in the paper, which have been widely used in hypersonic numerical simulations [46,49,56].

There are mainly two methods of measuring uncertainty in the coarse-grained uncertainty metrics; the first one is to compare the experimental data directly with the numerical simulation data, which we express in the form of $E(F)$ in the latter; and the second is to bring the data points of the experimental data into the fitted curve of the numerical simulation for comparison, which we express in the form

Table 3 Definitions of uncertainty metrics of corner cases of flat-plate cases [46]

Uncertainty metric type	Simplified equation
Weighted average comparison: $E(F_{\text{avg}})$	$E(F_{\text{avg}}) = \frac{\text{mean}(f_{\text{cfd}}) - \text{mean}(f_{\text{exp}})}{\text{mean}(f_{\text{exp}})} \times 100\%$
Maximum comparison: $E(F_{\text{max}})$	$E(F_{\text{max}}) = \frac{\max(f_{\text{cfd}}) - \max(f_{\text{exp}})}{\max(f_{\text{exp}})} \times 100\%$
Average of comparing points: $E(f_{\text{avg}})$	$E(f_{\text{avg}}) = \text{mean} \left(\left \frac{f_{\text{cfd}}(x_{\text{exp}}) - f_{\text{exp}}(x_{\text{exp}})}{f_{\text{exp}}(x_{\text{exp}})} \right \right) \times 100\%$
Maximum of comparing points: $E(f_{\text{max}})$	$E(f_{\text{max}}) = \max \left(\left \frac{f_{\text{cfd}}(x_{\text{exp}}) - f_{\text{exp}}(x_{\text{exp}})}{f_{\text{exp}}(x_{\text{exp}})} \right \right) \times 100\%$

of $E(f)$ in the latter. The specific uncertainty metrics involved are shown in Table 3.

Of these four metrics, $E(F_{\text{avg}})$ compares the error of the weighted average of the experimental and simulated data; but because it averages the data first, it tends to cancel out the errors. So, it can only be used to measure the overall level of error and is often used to measure the overall aerodynamic force of the vehicle control surfaces. $E(F_{\text{max}})$ compares the error of the maximum values of the experimental data and the simulated data, which has a more important application in engineering. For example, in the case of thermal protection, the error between the maximum values can be taken into account to better determine the amount of thermal protection material to be used; $E(f_{\text{max}})$ determines the maximum error at the same location by fitting a curve to the numerical simulation data, which is particularly effective when there are strong intermittent solutions or separation regions in the calculation, allowing the location of the maximum error transfer to be quickly identified. $E(f_{\text{avg}})$ determines the error between the experimental and simulated data at each location by curve fitting to the numerical simulation data, and a weighted average of all the error values is used, which is the most accurate reference value for the numerical simulation.

In summary, in the error analysis of the flat-plate cases, we will analyze the error by plotting the error profile and combining the coarse-grained uncertainty metrics $E(f_{\text{avg}})$ and $E(f_{\text{max}})$; whereas in the error analysis of the corner cases, we first divide the corner into a precompression region and a postcompression region, and we calculate the four uncertainty metrics in each of the two regions, respectively. Subsequently, $E(f_{\text{avg}})$ was selected as the reference to calculate the model error metrics E_{mod} , for which the specific expression is shown in Eq. (11), with $N(x)$ representing the number of data points:

$$E_{\text{mod}} = \frac{N(x_{\text{pre-exp}})}{N(x_{\text{exp}})} E(f_{\text{avg}}^{\text{pre}}) + \frac{N(x_{\text{post-exp}})}{N(x_{\text{exp}})} E(f_{\text{avg}}^{\text{post}}) \quad (11)$$

IV. Results and Discussions

The detail of all turbulence models in the latter part of the paper is introduced in Fig. 3. The numbers in the models denote the correction terms contained in Eq. (10).

A. Results of Hypersonic Flow over Flat Plate: Case M6Tw076

In the flat-plate cases, the accuracy of the model is judged by comparing the wall friction coefficient, the wall heat transfer coefficient, the velocity profile, and the temperature profile. The location of the profile comparison for this case is $Re_\tau \approx 450$.

As can be seen from Figs. 4 and 5, all SA models can be broadly divided into two echelons in terms of wall coefficients. All turbulence models containing β_4 (Shur et al. [38] compressibility correction) significantly reduce the predicted wall coefficients as compared to the original model or other compressible models; whereas other compressibility correction models increase the predicted wall coefficients as compared to the original model, with the β_1 term (conservation correction) bringing the most significant increase and the β_3, β_6 term (S  r  r et al. correction [41]) having a very small effect on the coefficients results. The presence of β_5 has a slight effect on the

	Models	Definition	Line Style
Incompressible Models	SA	/	--□--
	SA-COMP	Shur et al.'s [38] Correction	--◇--
Compressible Models (Single Correction)	SA-C5	SA-2012 [42] Correction	—□—
	SA-C4	Shur et al.'s [38] Correction	—□—
	SA-C3	SA-Catris-1999 [39] Correction	—□—
	SA-C1	Conservation [46] Correction	—□—
	SA-C36	Séror et al.'s [41] Correction	—◇—
	SA-C23	SA-Catris-2000 [40] Correction	—□—
Compressible Models (Multiple Corrections)	SA-C123	Conservation [46]+SA-Catris-2000 [40]	--◇--
	SA-C234	SA-Catris-2000 [40]+Shur et al.'s [38]	--◇--
	SA-C235	SA-Catris-2000 [40]+SA-2012 [42]	--△--
	SA-C236	SA-Catris-2000 [40]+Séror et al.'s [41]	--△--
	SA-C346	Séror et al.'s [41]+Shur et al.'s [38]	--◇--
	SA-C356	Séror et al.'s [41]+SA-2012 [42]	--◇--
	SA-C45	Shur et al.'s [38]+SA-2012 [42]	--△--

Fig. 3 Summary of SA models investigated in this paper.

coefficients but cannot be completely ignored as in some of the papers; in terms of the profile, the main errors in the velocity profile are concentrated in the viscous sublayer. However, the main errors in the temperature profile are concentrated in the log-law region, and there is no fixed regularity in the effect of each correction term on the profile results of the model.

Table 4 gives a detailed summary of the errors, with the maximum values noted, as well as the minimum values for the corresponding indicators. It can be seen that the SA model maintains an absolute advantage in the wall friction coefficient; the compressibility correction models (such as the SA-C36 model), which do not contain β_4 maintain a slight advantage in the wall heat transfer coefficient; the overall error difference in the profile is not too large, except for the maximum error value in the velocity profile, where the model with β_4 maintains a better advantage. Overall, most of the SA models perform well under this case, with some of the compressibility corrections reducing the errors in the wall heat transfer coefficients, although the effect of the corrections on the profile is not significant.

B. Results of Hypersonic Flow over Flat Plate: Case M6Tw025

This case is used to verify cold-wall effects on the individual turbulence models. The location of the profiles comparison for this case is $Re_\tau \approx 450$. As can be seen in Figs. 6 and 7, when the cold-wall effect is enhanced, the regularity of the wall coefficients for each turbulence model falls into roughly the same two echelons as for the normal wall: the turbulence model with β_4 decreases the predicted value significantly relative to the SA model, whereas the turbulence model without β_4 increases the predicted value slightly. The effect of each of the other correction terms is similar to that of the normal wall. The difference is that the SA model predictions of the wall coefficients deviate downward from the DNS data as the cold-wall effect increases, which also leads to an increase in the accuracy of some of the compressibility corrections due to the increased prediction values. There is some discrepancy in the profiles, where the presence of β_4 significantly reduces the accuracy of the velocity and temperature profiles in the lower half of the log-law region and viscous sublayer. There is no fixed regularity in the effect of each correction term in the upper part of the log-law region.

Table 5 details the error summary for the M6Tw025 case. As in the previous table, the maximum and minimum values for each indicator are also marked. It can be seen that all models without β_4 show some improvement over the incompressible models, and the SA-C1 model shows a definite advantage in the prediction of wall coefficients and profiles under the cold-wall effect. β_4 no longer has an optimizing effect on the maximum error in the velocity profile under the cold-

wall effect but rather exacerbates the increase in error. Overall, all correction models have different effects on the profiles and wall coefficients under the cold-wall effect, but the wall coefficients have the most significant correction effect.

C. Results of Hypersonic Flow over Flat Plate: Case M8Tw048

This case focuses on verifying the effect of increasing the Mach number on the predictions of the individual turbulence models. The location of the profiles comparison for this case is $Re_\tau \approx 480$. As can be seen in Figs. 8 and 9, the overall distribution of the individual correction terms remains the same as in the previous two cases in terms of wall coefficients, but increasing the Mach number further improves the predictions of the turbulence model relative to the DNS data. Therefore, it can be predicted that if the Mach number is increased again, the correction term containing the β_4 may have a more significant advantage. In the velocity profile, the β_4 correction continues to reduce the predictions significantly as compared to the other corrections in the lower half of the log-law region and the viscous sublayer, whereas the temperature profile does not show a significant error distribution regularity. Therefore, it can be seen that the overall effect of increasing the Mach number on each turbulence model is mainly on the increase of wall coefficients, with little effect on the profiles.

As can be seen from Table 6, the SA model maintains an absolute advantage in the prediction of the wall coefficients due to the high Mach number, which further improves the prediction of the turbulence model in this case. The overall difference between all models in terms of profile remains small.

D. Results of Hypersonic Flow over Compression Corner of 15 Degree Angle

In the three previous test cases, it can be found that the traditional incompressible SA model still predicts the wall coefficients and profiles well, even at high Mach numbers, except when the cold-wall effect increases significantly, where the partial compressibility correction model shows an absolute prediction advantage in terms of wall coefficients. This may be related to the fact that the flat-plate case itself does not have an overly pronounced strong intermittent solution or a region where the compressibility effect is evident. Therefore, in order to further explore the application of the compressibility correction term to the turbulence model at high Mach numbers, we further consider the 15 deg corner case and the 34 deg corner case, where the strong adverse pressure gradient and the high-density region can better reflect the compressibility effect.

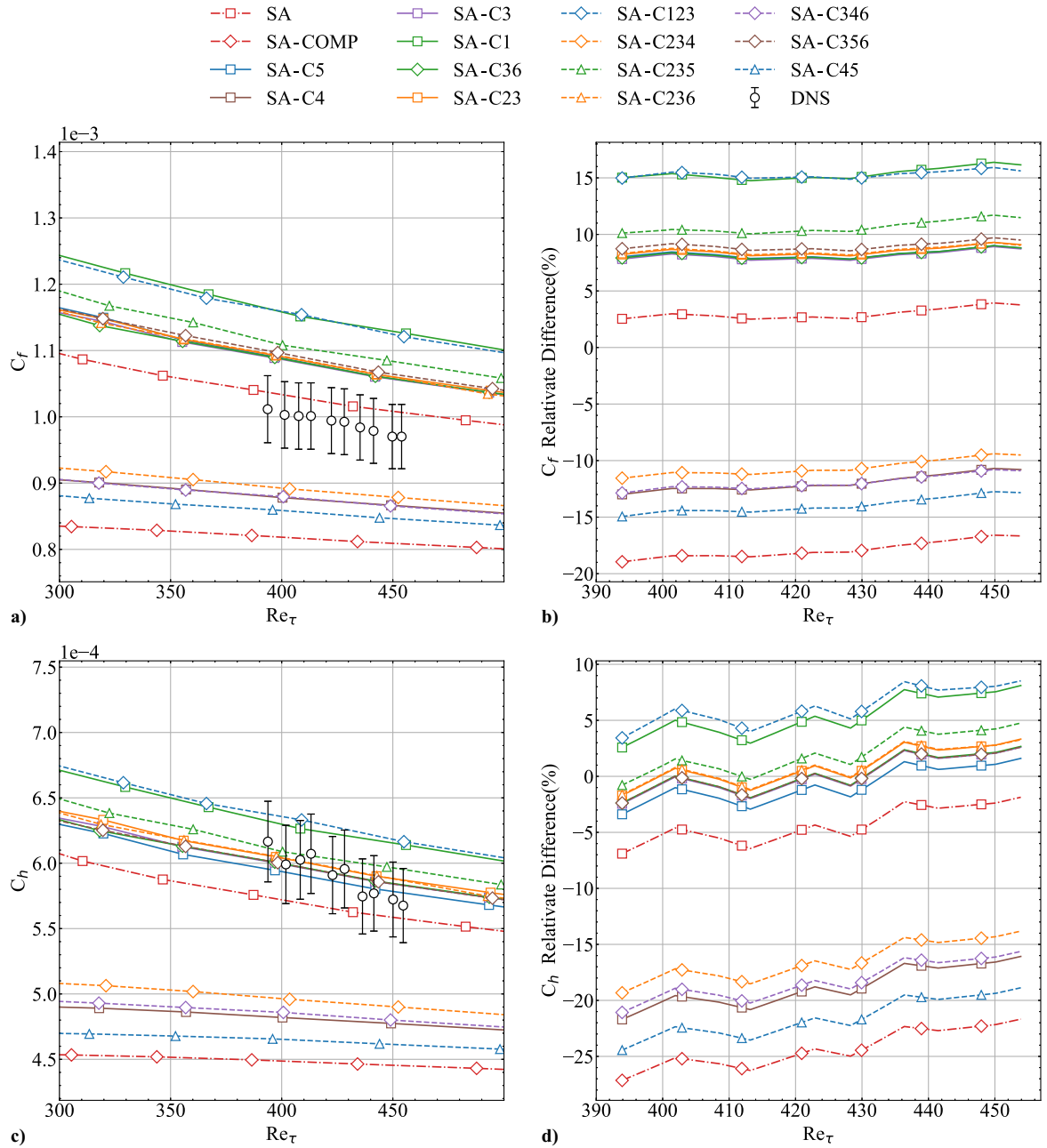


Fig. 4 Surface coefficients of M6Tw076: a) wall friction coefficient, b) error of friction coefficient, c) wall heat transfer coefficient, and d) error of heat transfer coefficient.

The pressure profile and heat flux profile are compared with those in the experimental data to check the accuracy of the predictions. Figure 10 shows the pressure and heat flux profiles for the 15 deg corner case. It is quite evident that the two conventional incompressible SA models do not correctly predict the large gradients of the pressure and heat flux at the start of the corner but instead show a more subdued transformation trend, which is not consistent with the actual physical phenomena; and there are false peaks in the postcompression region. In contrast, most of the compressibility correction terms are able to make some corrections relative to the incompressible model and correctly predict the large gradient changes at the beginning of the corner.

Looking further at Tables 7 and 8, it can be seen that in terms of pressure, All the compressible corrections have different optimization effects as compared to the incompressible model, except for SA-C1 and SA-C123, which perform worse in the postcompression region $E(f_{avg})$ metrics. By comparing the E_{mod} metrics, it can be seen that all models except the models with β_1 and β_4 terms have quite good correction effects. In terms of heat flux, all models significantly

reduce the prediction error, with the SA-C1 and SA-C123 models showing an impressive advantage in the postcompression region $E(f_{avg})$ metric, which further leads to the best performance in the E_{mod} metric.

With respect to the separation region, the incompressible SA model incorrectly predicts the point of separation earlier. The main influence between the different compressible correction models is the prediction of the reattachment point, where models with the β_1 term tend to delay the reattachment, whereas models with the β_4 term get an earlier reattachment point.

E. Results of Hypersonic Flow over Compression Corner of 34 Degree Angle

Figure 11 shows the pressure profile and heat flux profile for the 34 deg corner case. It can be seen that all the compressible correction models with β_4 are not shown in the figure as compared to the previous cases because all the compressible correction models with β_4 suffer from severe convergence problems in this case. This shows

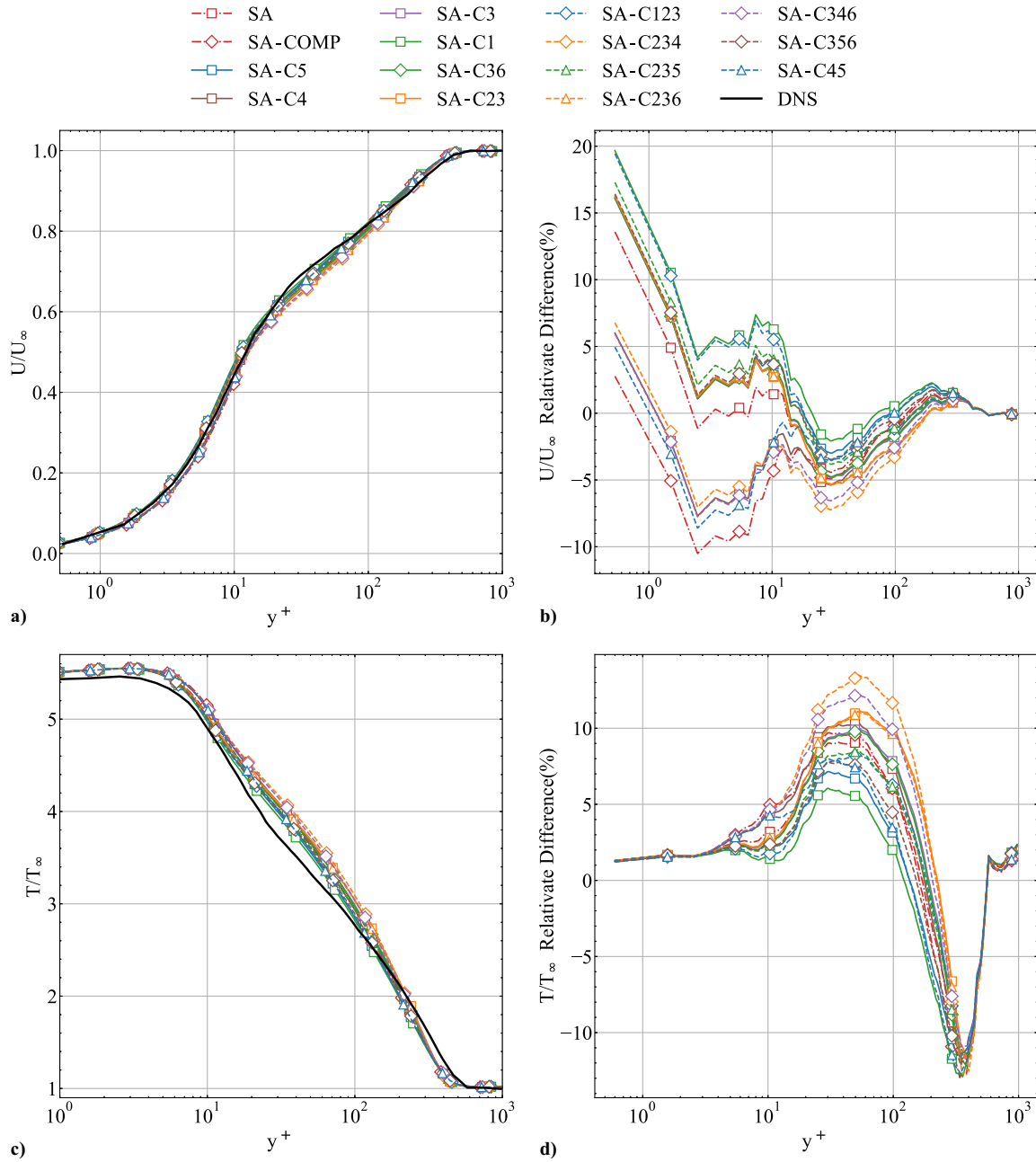


Fig. 5 Profiles of case M6Tw076: a) velocity profile, b) error of velocity profile, c) temperature profile, and d) error of temperature profile.

Table 4 Summary of errors for M6Tw076 case

	$C_f(f_{avg})$	$C_f(f_{max})$	$C_h(f_{avg})$	$C_h(f_{max})$	$U(f_{avg})$	$U(f_{max})$	$T(f_{avg})$	$T(f_{max})$
SA	2.99 ^b	3.94 ^b	4.35	-6.9	0.62	13.59	4.31	-11.64
SA-COMP	17.89 ^a	-18.95 ^a	24.33 ^a	-27.14 ^a	0.77	-10.51	4.61	-12.96
SA-C5	8.26	9	1.46	-3.39	0.63	16.12	4.29	-11.71
SA-C4	11.97	-12.99	18.8	-21.7	0.72	-7.7	4.43	-11.81
SA-C3	8.14	8.92	1.23	2.6 ^b	0.66	16.11	4.45	-11.85
SA-C1	15.36	16.38	5.36	8.1	0.72	19.67 ^a	4.59	-12.91
SA-C36	8.23	9.01	1.22 ^b	2.67	0.67	16.14	4.53	-12.24
SA-C23	8.53	9.3	1.3	3.33	0.71	16.28	4.56	-11.83
SA-C123	15.31	15.92	6.19	8.53	0.61	19.44	4.2 ^b	-11.42
SA-C234	10.62	-11.55	16.49	-19.32	0.9 ^a	-7.24 ^b	5.02 ^a	13.52 ^a
SA-C235	10.63	11.7	2.09	4.76	0.6 ^b	17.29	4.24	-11.39 ^b
SA-C236	8.6	9.29	1.32	3.26	0.71	16.3	4.6	-12.11
SA-C346	11.94	-12.87	18.25	-21.07	0.79	-7.76	4.49	-12.32
SA-C356	8.98	9.71	1.22	2.61	0.63	16.43	4.4	-12.15
SA-C45	13.97	-14.94	21.57	-24.44	0.7	-8.6	4.51	-12.74

^aMaximum values.

^bCorresponding minimum values.

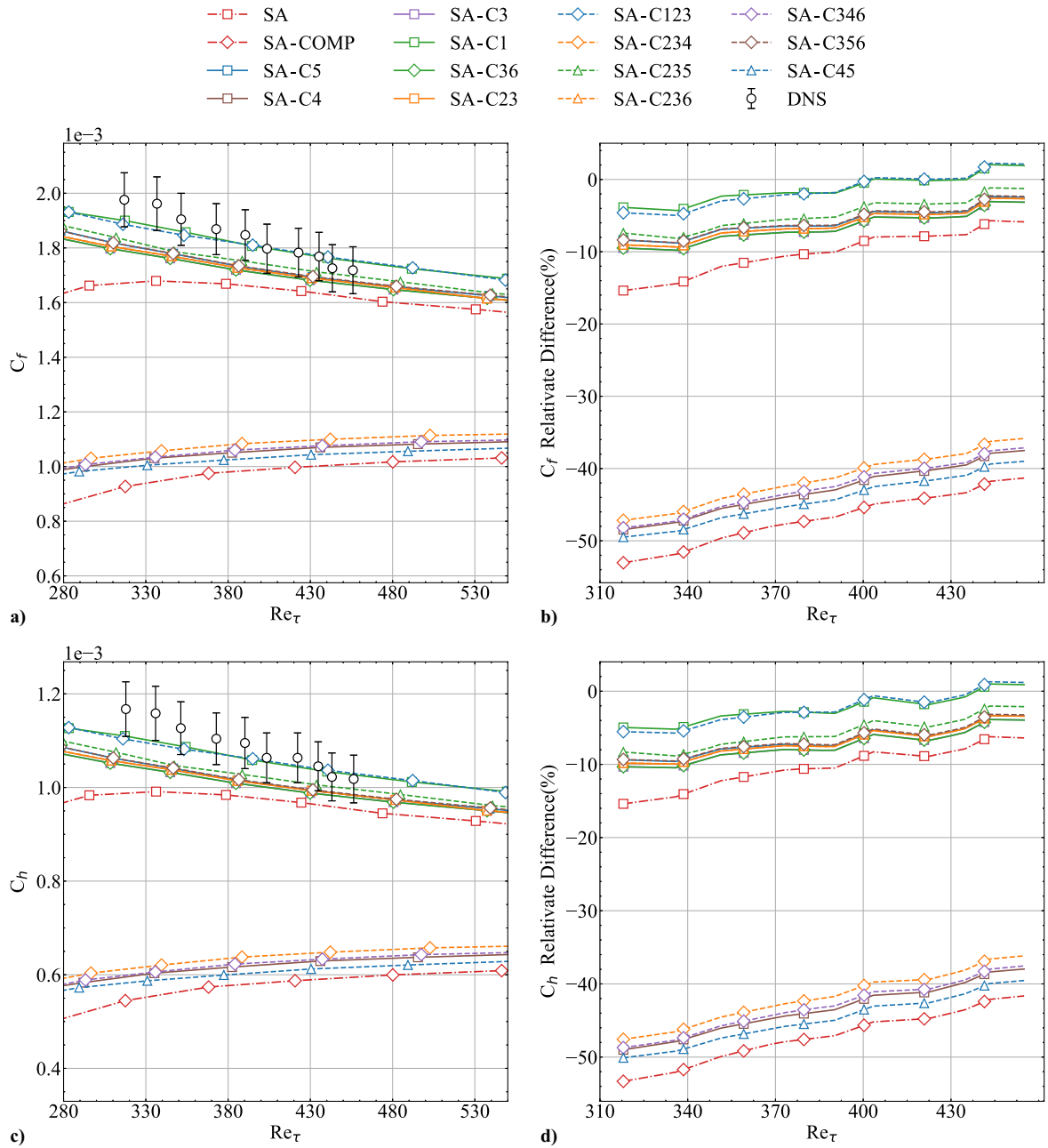


Fig. 6 Surface coefficients of M6Tw025: a) wall friction coefficient, b) error of friction coefficient, c) wall heat transfer coefficient, and d) error of heat transfer coefficient.

that β_4 does not fit cases where the compression effect is very strong. In addition, it can be seen that the incompressible model significantly overestimates the gradients of heat flux and pressure profiles in the transition area, whereas the compressible correction model with β_1 significantly underestimates the gradients of the heat flux and pressure profiles in the transition area, resulting in poor predictions for both types of models for the strong compressibility case.

It can be seen from Tables 9 and 10 that, in terms of pressure, no single model has a very clear predictive advantage in terms of all metrics; of these, the compressibility correction models with β_1 predict extremely poorly, with errors even exceeding 100%. However, when we focus on the $E(f_{avg})$ metrics in the postcompression region, Catris correction models [39,40] and S  ror et al. correction model [41] improve the prediction of the pressure profile by about 60% as compared to the incompressible model. The main reason why their E_{mod} are still worse is that they overpredicted the start of the separation region in the precompression region. Considering that the

absolute error in the precompression region is still very small, we believe that these two types of models are still the best methods for predicting pressure profile, rather than the SA-C235 model with the smallest E_{mod} . Surprisingly, the original SA model predicts the $E(F_{avg})$ and $E(F_{max})$ metrics relatively well under strong compressibility effects in the postcompression region, which may be one of the reasons why the original SA model is still sufficient for engineering in the overall prediction of hypersonic vehicle aerodynamics. In terms of heat flux, all correction models without β_1 have some correction effect relative to the incompressible model, with the SA-C23, SA-C236, and SA-C36 models having the best correction effects, with errors below 25% at the E_{mod} metric.

With respect to the separation region, as the compressibility effect increases, the prediction accuracy of the separation region decreases for all models. Among them, the incompressible models and models with β_3 lead to an earlier reattachment. Models with β_1 tend to delay the reattachment, whereas Catris [39,40] and S  ror et al. [41] correction models result in an earlier separation.

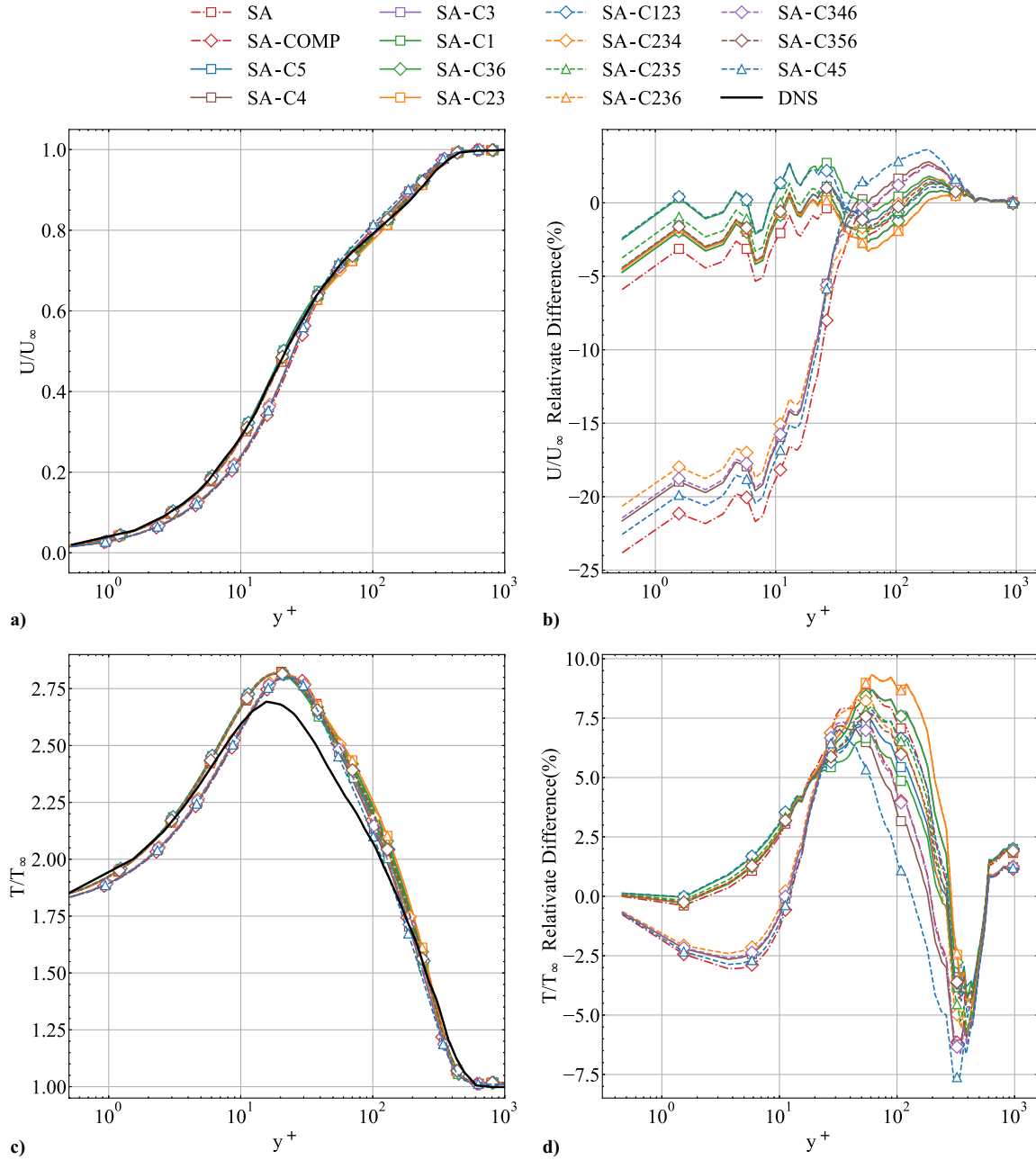


Fig. 7 Profiles of case M6Tw025: a) velocity profile, b) error of velocity profile, c) temperature profile, and d) error of temperature profile.

Table 5 Summary of errors for M6Tw025 case

	$C_f(f_{avg})$	$C_f(f_{max})$	$C_h(f_{avg})$	$C_h(f_{max})$	$U(f_{avg})$	$U(f_{max})$	$T(f_{avg})$	$T(f_{max})$
SA	10.11	-15.37	10.44	-15.39	0.51	-5.9	2.9	8.68
SA-COMP	46.93 ^a	-53.04 ^a	47.23 ^a	-53.31 ^a	1.14	-23.82 ^a	2.56	7.96
SA-C5	5.86	-8.79	6.81	-9.61	0.43	-4.46	2.51	7.35
SA-C4	42.98	-48.45	43.48	-49	1.09	-21.65	2.51	7.18
SA-C3	6.69	-9.79	7.5	-10.45	0.43	-4.73	2.94	8.69
SA-C1	1.81 ^b	-4.29 ^b	2.6 ^b	-5.19 ^b	0.48	2.72 ^b	2.50 ^b	6.73 ^b
SA-C36	6.71	-9.8	7.51	-10.47	0.42 ^b	-4.73	2.93	8.67
SA-C23	6.24	-9.36	6.99	-9.96	0.46	-4.54	3.22 ^a	9.33 ^a
SA-C123	2.1	-4.98	2.73	-5.73	0.43	2.74	2.9	7.83
SA-C234	41.46	-47.16	41.82	-47.57	0.9	-20.64	2.66	8.28
SA-C235	4.88	-8.16	5.74	-8.85	0.48	-3.75	2.96	8
SA-C236	6.22	-9.34	6.98	-9.95	0.45	-4.53	3.2	9.31
SA-C346	42.64	-48.21	43.09	-48.7	1.04	-21.44	2.6	7.38
SA-C356	5.78	-8.76	6.7	-9.54	0.44	-4.41	2.67	7.69
SA-C45	44.31	-49.49	44.89	-50.1	1.34 ^a	-22.56	2.71	-7.64

^aMaximum values.

^bCorresponding minimum values.

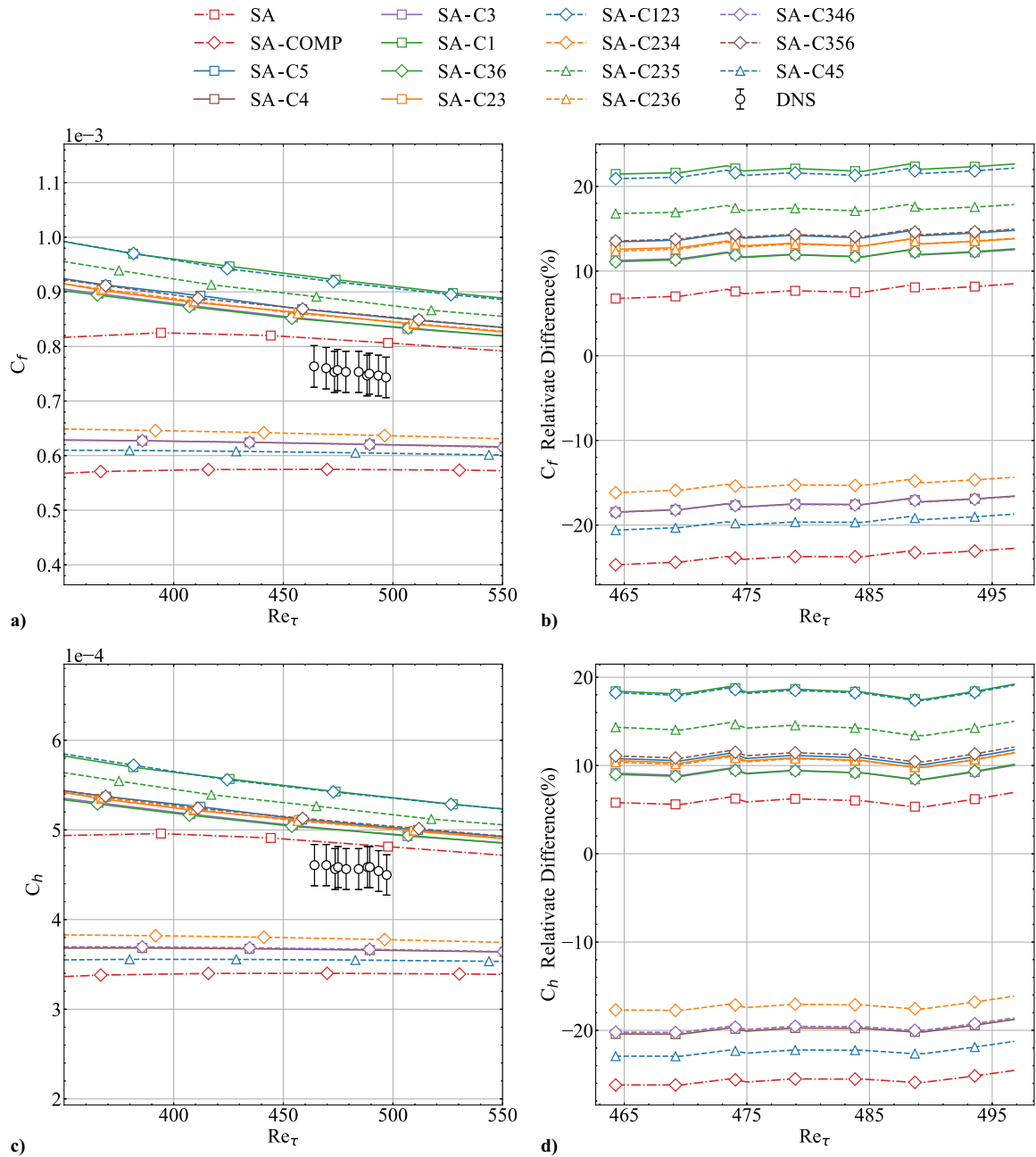


Fig. 8 Surface coefficients of M8Tw048: a) wall friction coefficient, b) error of friction coefficient, c) wall heat transfer coefficient, and d) error of heat transfer coefficient.

V. Conclusions

The SA model is one of the most commonly used turbulence models in the engineering field, and its compressibility corrections have not been measured in a comprehensive and standard way for hypersonic performance. In this paper, first, the compressibility corrections applied to the SA model in recent years are summarized, and their effectiveness is examined in three flat-plate flow cases with different Mach numbers and wall temperatures, as well as two hypersonic corner flow cases with two corner angles.

The results show that the traditional incompressible SA model can measure the wall coefficient and profile more accurately when the cold-wall effect is not strong in the flat-plate case. As the cold-wall effect increases, the original SA model gives an underprediction of the wall coefficient; and the necessity of the compressibility correction term gradually emerges. Among them, the SA-C1 and SA-C123 models show a remarkable advantage in terms of wall coefficients, thus demonstrating the need for β_1 under the cold-wall effect. As the Mach number increases, the SA model predictions for the wall

coefficients become progressively higher; given that β_4 reduces the wall coefficient predictions significantly, it is to be expected that the β_4 term may maintain some advantage at very high Mach numbers. In terms of profile prediction, all SA models are able to measure the profile more accurately; and most of the compressibility correction terms have some correction effect on the profile under the cold-wall effect: except the β_4 term, leading to an increase in the prediction error for the lower half of the log-law region and the viscous sublayer under the cold-wall effect, as well as an increase in the maximum error value for the velocity profile. However, the overall correction effect on profile prediction is not significant. For other SA correction terms, the S  r et al. correction model (β_3, β_6) [41] presents a very small effect on the results of the hypersonic flat-plate case, Catris and Aupoix correction models (β_2, β_3) [40] increases the predicted value of the wall coefficient by a small amount, and the β_5 term has a small effect on the wall coefficient but is still not negligible.

In the corner case, the incompressible model performs very poorly due to the strong adverse pressure gradient and the high-density

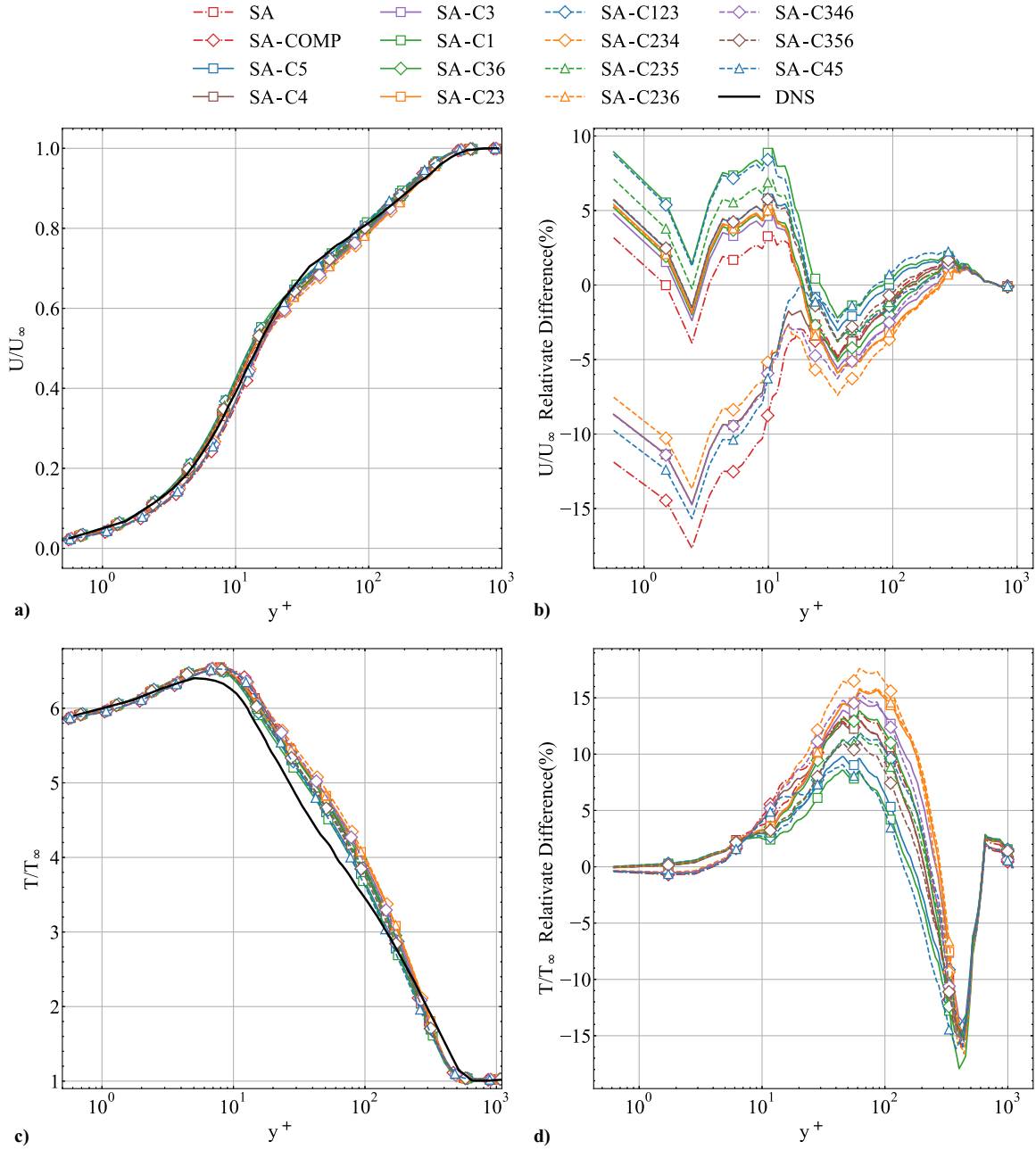


Fig. 9 Profiles of case M8Tw048: a) velocity profile, b) error of velocity profile, c) temperature profile, and d) error of temperature profile.

Table 6 Summary of errors for M8Tw048 case

	$C_f(f_{avg})$	$C_f(f_{max})$	$C_h(f_{avg})$	$C_h(f_{max})$	$U(f_{avg})$	$U(f_{max})$	$T(f_{avg})$	$T(f_{max})$
SA	7.63 ^b	8.52 ^b	5.95 ^b	6.96 ^b	0.75	-4.8 ^b	5.31	-15.37
SA-COMP	23.72 ^a	-24.69 ^a	25.66 ^a	-26.19 ^a	0.95	-17.67 ^a	5.21	-16.04
SA-C5	14.13	14.81	10.83	11.77	0.78	6.07	4.95 ^b	-14.07 ^b
SA-C4	17.53	-18.46	19.91	-20.44	0.87	-14.71	5.09	-15.21
SA-C3	11.89	12.52	9.14	10.02	0.81	-5.63	5.38	14.8
SA-C1	22.03	22.65	18.32	19.21	0.86	9.17	5.15	-15.22
SA-C36	11.86	12.63	9.12	10.13	0.88	5.19	5.99	-17.92 ^a
SA-C23	13.19	13.82	10.54	11.42	0.92	-5.9	5.92	15.71
SA-C123	21.52	22.16	18.18	19.1	0.73 ^b	8.77	5.15	-15.05
SA-C234	15.26	-16.17	17.24	-17.75	1.12 ^a	-13.67	6.04 ^a	17.6
SA-C235	17.32	17.88	14.22	15.02	0.74	7.11	5.25	-15.59
SA-C236	13.11	13.88	10.47	11.48	0.92	-5.96	5.84	15.87
SA-C346	17.57	-18.49	19.72	-20.24	0.98	-14.75	5.51	-15.98
SA-C356	14.25	14.96	11.14	12.1	0.76	5.97	5.12	-14.75
SA-C45	19.65	-20.6	22.39	-22.94	0.96	-15.69	5.24	-16.16

^aMaximum values.

^bCorresponding minimum values.

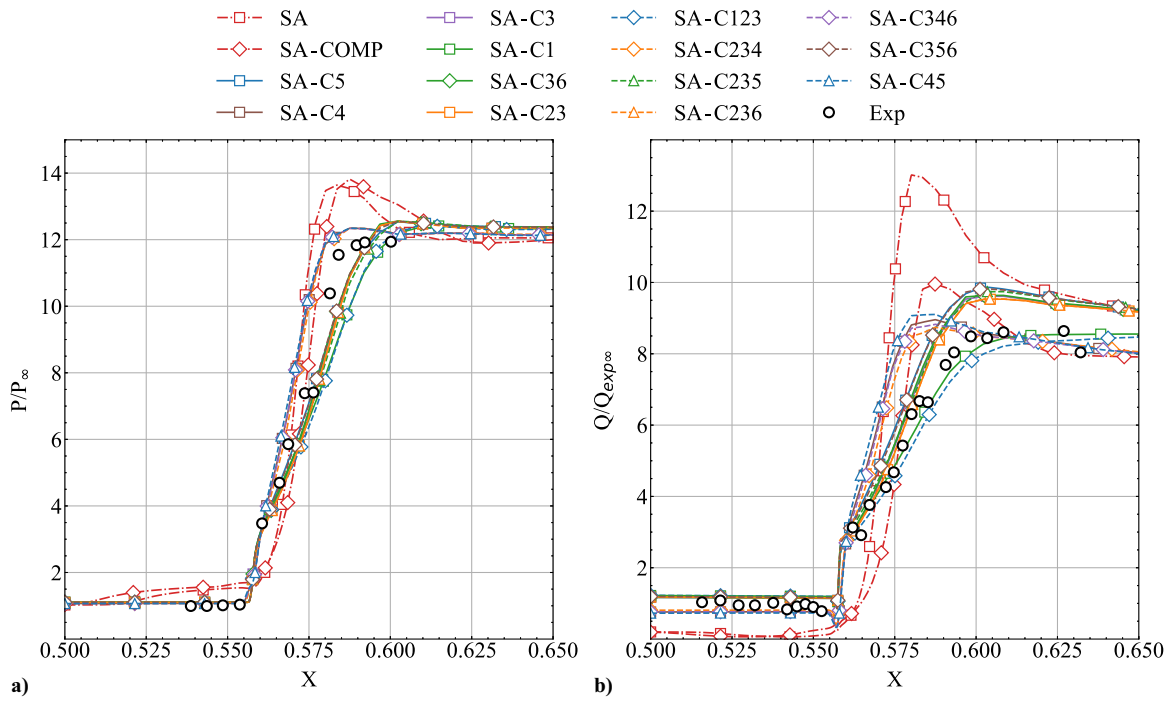


Fig. 10 Profiles of 15 deg corner case: a) pressure profile, and b) heat flux profile.

Table 7 Summary of pressure profile errors for 15 deg corner case (100%)^a

	$E(F_{avg})$		$E(F_{max})$		$E(f_{avg})$		E_{mod}	$E(f_{max})$	
	Pre	Post	Pre	Post	Pre	Post		Pre	Post
SA	50.7	-41	53.17	14.26	48.21	25.7 ^b	32.13 ^b	49.42	63.26 ^b
SA-COMP	64.4 ^b	-44.65	75.06 ^b	15.77 ^b	58.22 ^b	21.26	31.82	62.8 ^b	28.68
SA-C5	13.6	-42.77	42.23	2.37	9.42	5.91 ^c	6.91 ^c	11.13	4.01
SA-C4	20.42	-34.92	55.27	3.39	6.05	14.55	12.12	7.64	44.57
SA-C3	13.49 ^c	-44.08	41.92 ^c	4.06	9.24	7.84	8.24	10.92	4.7
SA-C1	14.82	-45.25	47.91	-1.77	9.69	10.21	10.06	11.39	0.67 ^c
SA-C36	13.52	-43.82	42.07	4.43	9.24	7.35	7.89	10.94	4.9
SA-C23	13.59	-44.6	42	4.02	9.38	8.89	9.03	11.07	4.73
SA-C123	14.76	-46.11 ^b	47.55	-0.86 ^c	9.72	11.84	11.23	11.41	1.1
SA-C234	18.5	-35.99	52.9	3.4	6.43	12.88	11.04	8.05	42.77
SA-C235	13.62	-44.31	41.95	1.79	9.58	8.59	8.87	11.28	3.89
SA-C236	13.59	-44.6	42.03	4.01	9.38	8.89	9.03	11.06	4.72
SA-C346	20.06	-35.01	54.67	3.37	6.07	14.82	12.32	7.66	45.74
SA-C356	13.62	-42.91	42.2	3.45	9.45	6.12	7.07	11.16	4.51
SA-C45	22.11	-33.98 ^c	57.13	3.43	5.82 ^c	16.05	13.13	7.34 ^c	46.76

^aPre denotes precompression, and post denotes postcompression.

^bMaximum values.

^cCorresponding minimum values.

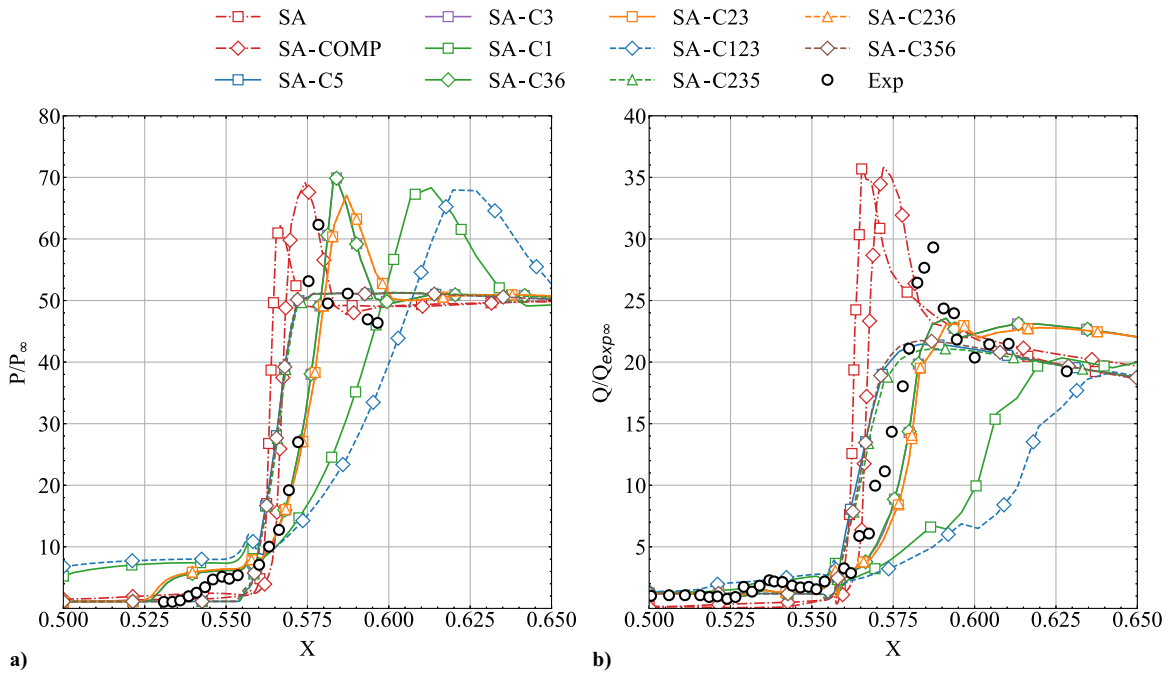
region of the corner case itself, where there are significant compressibility effects. Almost all of the compressibility corrections optimize the pressure and heat flux profiles, but the prediction error increases with an increasing corner angle (increasing compression effects), with the correction model containing β_4 showing serious convergence problems for the larger corner case and the correction model containing β_1 showing serious prediction errors for the larger corner case. In terms of the pressure profile, when the compression is weak (small corner angle), all of the modified models predict quite accurately, with the SA-C3, SA-C36, and SA-C5 models all maintaining the E_{mod} metric below 10%. As the compressibility effect increases, most of the compressible models' corrections to the pressure profile weaken, whereas the models containing Catris corrections [39,40] and Séror et al. correction [41] maintain a very significant predictive advantage in the $E(f_{avg})$ metrics in the postcompression region. In terms of heat flux profiles, the SA-C123 and SA-C1 models perform best in the small-angle cornering case. However, as the angle of the corner increases, the modified model with the β_1 term

significantly underestimates the gradient of the disturbance area; whereas Catris and Aupoix model [40], and Séror et al. [41] models (such as SA-C23, SA-C236, and SA-C36) perform more accurately, greatly improving the accuracy of the heat flux predictions. Combining the results of previous evaluations of the β_1 term in flat-plate cases and the small-angle corners, we can find that the conclusions of Gnoffo et al. [49] and Waligura et al. [46] are not entirely accurate, and that the β_1 term can indeed be detrimental to the prediction of the SA model under strong compression effects; but, it still provides a good improvement in the prediction of heat flux when the cold-wall effect is strong or when the compression effect is weak.

For the compressible models, two source terms relating to the velocity gradient are added to the β_1 and β_4 terms, respectively. β_1 greatly increases the predicted value of the turbulent viscosity, which results in additional turbulence in the flowfield. It further changes the flow structure near the wall, leading to a significant increase in the predicted values of the wall friction and heat flux for the flat-plate cases. For β_4 , the additional source term greatly reduces the predicted

Table 8 Summary of heat flux profile for 15 deg corner case (100%)

	$E(F_{avg})$		$E(F_{max})$		$E(f_{avg})$		E_{mod}	$E(f_{max})$	
	Pre	Post	Pre	Post	Pre	Post		Pre	Post
SA	-84.92 ^a	-39.97	-72.69 ^a	50.69 ^a	89.41 ^a	60.8 ^a	71.8 ^a	-81.96 ^a	120.37 ^a
SA-COMP	-70.38	-58.91 ^a	-60.25	15.28	85	28.32	50.12	-65.32	48.67
SA-C5	19.43	-33.42	8.32	12.14	25.29	17.39	20.43	48.9	29.24
SA-C4	-30.7	-33.23	-29.07	3.69	18.19	26.4	23.24	-1.92	65.19
SA-C3	17.3 ^b	-37.5	6.4 ^b	10.37	23.07	11.6	16.01	46.39	18.47
SA-C1	21.92	-42.57	11.28	-8.06	28.72	4.86 ^b	14.04 ^b	53.02	15.65
SA-C36	17.32	-37.14	6.5	11.07	23.19	12.31	16.49	46.46	19.72
SA-C23	18.7	-38.64	7.63	8.88	24.5	9.71	15.4	48.07	15.9
SA-C123	22.46	-44.79	11.69	-10.74	29.19	7.14	15.62	53.6	9.69 ^b
SA-C234	-24.07	-35.86	-24.71	1.24 ^b	13.85 ^b	20.97	18.23	4.12	54.34
SA-C235	21.97	-36.19	10.49	9.43	27.8	12.3	18.26	51.88	21.95
SA-C236	18.67	-38.61	7.59	8.92	24.46	9.76	15.41	48.02	15.92
SA-C346	-30.11	-34.38	-28.77	2.2	17.87	24.65	22.04	-1.49 ^b	63.58
SA-C356	19.67	-34.15	8.53	12.63	25.53	16.34	19.87	49.21	26.21
SA-C45	-35.53	-31.11 ^b	-32.3	5.42	21.9	30.75	27.35	-6.4	73.86

^aMaximum values.^bCorresponding minimum values.**Fig. 11** Profiles of 34 deg corner case: a) pressure profile, and b) heat flux profile.**Table 9** Summary of pressure profile errors for 34 deg corner case (100%)

	$E(F_{avg})$		$E(F_{max})$		$E(f_{avg})$		E_{mod}	$E(f_{max})$	
	Pre	Post	Pre	Post	Pre	Post		Pre	Post
SA	-39.68	-26.4 ^b	-51.71	-0.24 ^b	46.95 ^b	85.6 ^a	67.2	39.25	382.72 ^a
SA-COMP	-22.19 ^b	-37.04	-45.07	11	55.91	52.86	54.31	111.55	196.06
SA-C5	-41.68	-31.58	-25.69	-17.9	50.01	49.81	49.91	5.46 ^b	147.38
SA-C3	101.96	-43.37	51.58	13.78	135.17	13.79	71.59	340.69	26.77
SA-C1	171.04	-59.87	104.74	-26.66	243.27	33.2	133.23	590.8	15.35 ^b
SA-C36	101.97	-43.35	51.68	13.72	134.79	13.73 ^b	71.38	339.53	26.61
SA-C23	112.47	-43.93	59.23	7.76	151.84	16.85	81.13	378.13	30.46
SA-C123	198.53 ^a	-62.19 ^a	124.42 ^a	-44.91 ^a	271.53 ^a	41.27	150.92 ^a	645.99 ^a	26.91
SA-C235	-39.7	-33.1	-23.28	-17.85	49.94	45.61	47.67 ^b	5.68	132.52
SA-C236	112.5	-43.92	59.37	7.7	151.64	16.82	81.02	377.57	30.39
SA-C356	-34.32	-32.24	-18.82 ^b	-17.86	49.4	48.19	48.77	5.46	146.08

^aMaximum values.^bCorresponding minimum values.

Table 10 Summary of heat flux profile for 34 deg corner case (100%)

	$E(F_{avg})$		$E(F_{max})$		$E(f_{avg})$		E_{mod}	$E(f_{max})$	
	Pre	Post	Pre	Post	Pre	Post		Pre	Post
SA	-58.94	-14.99 ^b	-46.94	22.58	87.34 ^a	92.83 ^a	89.95 ^a	-64.01	472.58 ^a
SA-COMP	-54.08	-36.44	-50.52	22.26	74.97	59.9	67.81	-56.56	263.36
SA-C5	-17.71	-39.12	-13.09 ^b	-26.85	29.75	42.04	35.59	55.65	186.45
SA-C3	28.82	-65.08	47.05	-19.59 ^b	22.63	22.58	22.61	48.33	18.69
SA-C1	87.88	-81.96	75.18 ^a	-73.49	30.53	56.62	42.92	91.18	5.32 ^b
SA-C36	29.43	-64.91	47.43	-19.63	22.69	22.3 ^b	22.5 ^b	48.62	18.73
SA-C23	34.12	-66.63	51.02	-21.08	20.74	25.35	22.93	42.63 ^b	17.95
SA-C123	99.27 ^a	-83.72 ^a	73.02	-76.58 ^a	49.52	64.84	56.8	163.89 ^a	-12.84
SA-C235	-17.58 ^b	-43.04	-14.11	-28.11	30.49	35.04	32.65	58.98	142.17
SA-C236	34.67	-66.54	51.45	-21.05	20.71 ^b	25.35	22.91	42.92	18
SA-C356	-18.04	-41.06	-14.72	-25.8	30.4	38.1	34.06	55.42	148.16

^aMaximum values.^bCorresponding minimum values.

value of the turbulent viscosity, with subsequent effects on the flat-plate boundary layer completely opposite to those of β_1 . Besides, one can see from the velocity profiles of the flat-plate cases that the β_4 term results in a significant reduction of the velocity in the inner layer. The lower velocity makes the flow more affected by the adverse pressure gradient, which in turn leads to an earlier reattachment point in corner cases. When the adverse pressure gradient is further increased, the more complex flow structure may lead to problems with the convergence of the model. The β_1 term, on the other hand, further increases the inner boundary-layer velocity, which greatly delays the appearance of the reattachment point. Other corrections such as β_2 , β_3 , and β_6 are diffusion terms related to the c_{b2} term; whereas β_5 is a diffusion term unrelated to c_{b2} . The corrective effect of these diffusion terms, however, is not as strong as that brought by the source term because they primarily affect the transfer of turbulent kinetic energy in the flowfield owing to changes in velocity and the turbulent kinetic energy gradient.

In summary, this paper provides a comprehensive assessment of the application of compressibility corrections to the SA model for high-Mach-number flows. Considering that the RANS model is still the most widely used method in the engineering field, the evaluation of the specific effects of each correction term in this paper can also provide reference data for subsequent researchers and engineers in selecting hypersonic turbulence models to further improve the accuracy of the SA model. However, the evaluation of each correction term in this work is not comprehensive enough, and the coarse-grained uncertainty metrics are not detailed and accurate enough. It is expected that the assessment will be carried out under more test cases in the future with detailed uncertainty quantification.

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